

1. Why is the policy in the case of Mountain Car time-dependent? In other words, why does the value of a given state depend on the current time step?

The policy in Mountain car is time-dependent because it's a finite-horizon problem. In Mountain Car you planned for only $T = 150$ steps, so the number of steps left changes how much future reward a state can still yield. As time runs out, there's less opportunity to recover or build momentum, so the optimal action may change. Mathematically, the value function is $V_t(s)$ so it depends on the current step t —so the resulting policy $\pi_t(s)$ is time-dependent.

2. What is the trade-off of having more states in your MDP? Think about how closely the MDP approximates the simulator.

Having more states means a finer discretization of the environment. That gives a closer approximation of the true behavior more accurately. The trade-off is Better accuracy (less “state aliasing,” smoother and more realistic policies) at a Much higher cost (exponentially more state–action pairs $\rightarrow$ more samples, slower model building, larger transition tables, and heavier computation for DP/VI.) In short, more states improve realism but greatly increase computation and sampling cost.